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Stick some more acknowledgments here.

Abstract

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Contents

1 Introduction **1**

1.1 Importance and Applications of Soft Robotics 1

1.2 Types of Self-Healing Polymers 3

1.3 Comparison of Different Self-Healing Pathways 5

1.3.1 Using a Diels-Alder Addition as a Self-Healing Pathway 6

1.3.2 Using a Thiol-Michael Addition as a Self-Healing Pathway 7

1.3.3 Using a Pi-Stacking Approach as a Self-Healing Pathway 7

1.4 Applications of Self-Healing Polymers in Soft Robotics 8

1.4.1 State of the Art Self-Healing Soft Robots 8

1.4.2 Requirements for Self-Healing Polymers in Soft Robotics 9

1.5 Limitations with Current Materials for Soft Robotics 9

2 Project Aims and Objectives **10**

3 Synthesis of a Self-Healing Polymer **11**

3.1 Results 11

3.1.1 Hydrolysis of the Polyimide 11

3.1.2 Spectrophotometry 13

3.1.3 Cross-linking of the Polyimide 14

3.1.4 Moulding of Polyamic Acid 17

3.1.5 Catalytic Imidisation of polyimide 17

3.2 Experimental 18

3.2.1 Equipment 18

3.2.2 The Hydrolysis of a Polyimide 18

3.2.3 Spectrophotometry as a Determination Method for Hydrolysis . . 19

3.2.4 Crosslinking of Polyamic acid 20

3.2.5 Crosslinking Spectral Data 20

3.2.6 Catalytic Imidisation of Polyamic Acid 21

4 Construction of a Bellows Based Actuator 22

4.1 Design 22

4.1.1 Construction of a Folded Protective Sleeve 22

4.1.2 Complete Assembly 24

4.2 Methodology 25

4.2.1 Stiffness Characterisation 25

4.2.2 Temperature Stability of the Polyimide Sleeve 26

4.2.3 Workspace Characterisation 26

4.2.4 Moulding of polyamic acid 27

4.2.5 Moulding of Polyamic Acid 27

4.3 Results 27

4.3.1 Stiffness Characterisation 27

4.3.2 Temperature Stability of the Bellows 28

4.3.3 Workspace Characterisation 29

4.3.4 Navigation 29

4.3.5 Clinical Case Study: Drug Delivery in Stomach 29

List of Figures

1.1 A continuum device sustaining damage before self-healing and regaining full functionality using a modified image from[1] 2

1.2 a) The polymer chain with a repeating chemical sub-unit. b) The mechanism of damage and repair for a self-healing polymer 4

1.3 a) A Diels-Alder reaction between two molecules of cylopentadiene. b) A Diels-Alder reaction between a furan and maleimide. c) A Thiol-Michael Addition between a thiol and a conjugated alkene. 6

4.1 A) The Kapton sleeve is placed into an oven for an hour and then placed onto the stiffness characterisation experimental setup. B) The initial profile of the actuator before heating. C) The profile after 3 cycles in the oven. D) A plot comparing the average force vs. displacement across all cycles 28

List of Tables

3.1 ATR-FTIR absorbance ratios and relative imide surface content for various samples in comparison to Wu (2017)[2] 13

3.2 table caption 19

3.3 table caption 19

3.4 table caption 20

Chapter 1

Introduction

1.1 Importance and Applications of Soft Robotics

Soft robots are a type of robot known for their use of soft materials, compliant nature and near infinite degrees of freedom . Degrees of freedom (DOF) is a term related to the number of independent movements a robot can make in 3D space.[3] Soft robots are usually nature inspired such as "vine-like" tip growing robots,[4,5] "snake like" continuum robots[6,7] and "hand-like" soft grippers.[8,9] The advantages of using a body that is highly deformable and with near-infinite DOF means these devices can navigate tortuous environments with ease.

This report focuses on continuum robots, often characterised by their long flexible bodies, commonly used for navigation and delivery or retrieval of objects in constrained environments.[10] Continuum robots can be classified into "continuous backbone" and "segmented back-bone" designs.[10] Continuous backbone devices are made from a soft and compliant material, in this report we will call these "soft-bodied", whereas, a segmented backbone refers to those that are made from hard, inflexible segments that are flexibly attached to achieve high curvatures, referred to here as "hard-bodied".

Hard-bodied continuum robots benefit from stiffer materials allowing a greater ease of

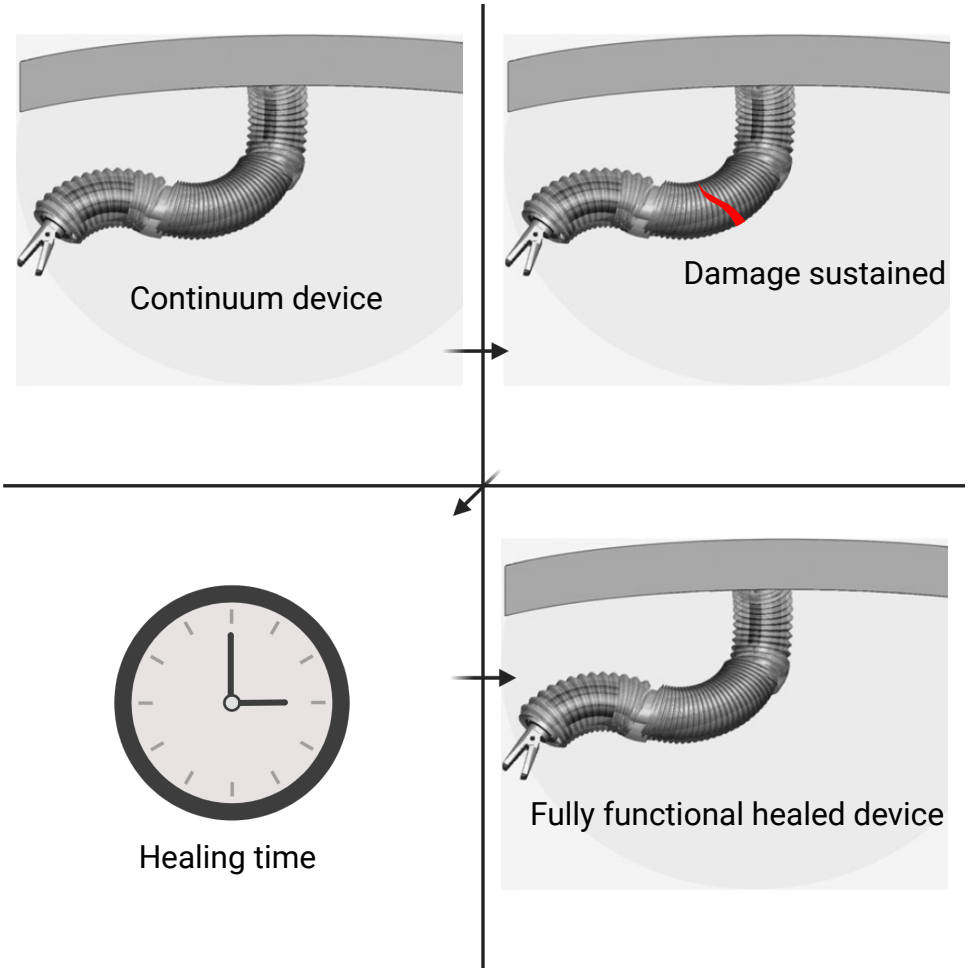


Figure 1.1: A continuum device sustaining damage before self-healing and regaining full functionality using a modified image from[1]

control[10] which can improve dexterity. However, they cannot deform and are more likely to impart a greater force on the surrounding environment.[11] Hard continuum robots are often tendon driven, whereby the segments are joined by wires that can be pulled to give specific movements. Whereas soft bodied continuum robots can achieve high curvatures and low forces on surroundings,[12] but are often harder to control due to their methods of actuation[10] and can suffer from material failure.[13] Soft continuum robots commonly take the form of an inflated body, these can be controlled in many ways such as varying pressure,[4] strain limiting a specific region[14,15] or magnetic actuation.[5,11] Pressurised systems that rely on fluids need a constant seal to maintain function as opposed to a segmented tendon driven approach that can be partially damaged and still actuatable. This leads the development of resilient materials such as self-healing polymers that can be applied to soft robotics.

1.2 Types of Self-Healing Polymers

Polymers are a type of molecule that consists of several repeating sub-units, these form chains that are combined many times over to form materials. Self-healing (S-H) polymers are a type of polymer that can heal damage.[16] This occurs through reactions and interactions between multiple polymer chains to reconnect chains that have been shifted apart through damage. The process of damaging or cutting a polymer does not break bonds in the polymer backbone but instead can break interchain interactions called cross-linking, allowing the chains to drift apart, reforming the cross-linking interactions will "heal" the material.[17]

S-H polymers can be classified into intrinsic or extrinsic healing, and autonomous or non-autonomous.[18] Materials with an intrinsic healing capacity involve a healing mechanism through chemical bonds or interactions that are continuously present in the polymer, rather than extrinsic healing, such as capsules of "healing" material that are broken through damage and causing the material to reform.[19] Autonomy refers to the healing

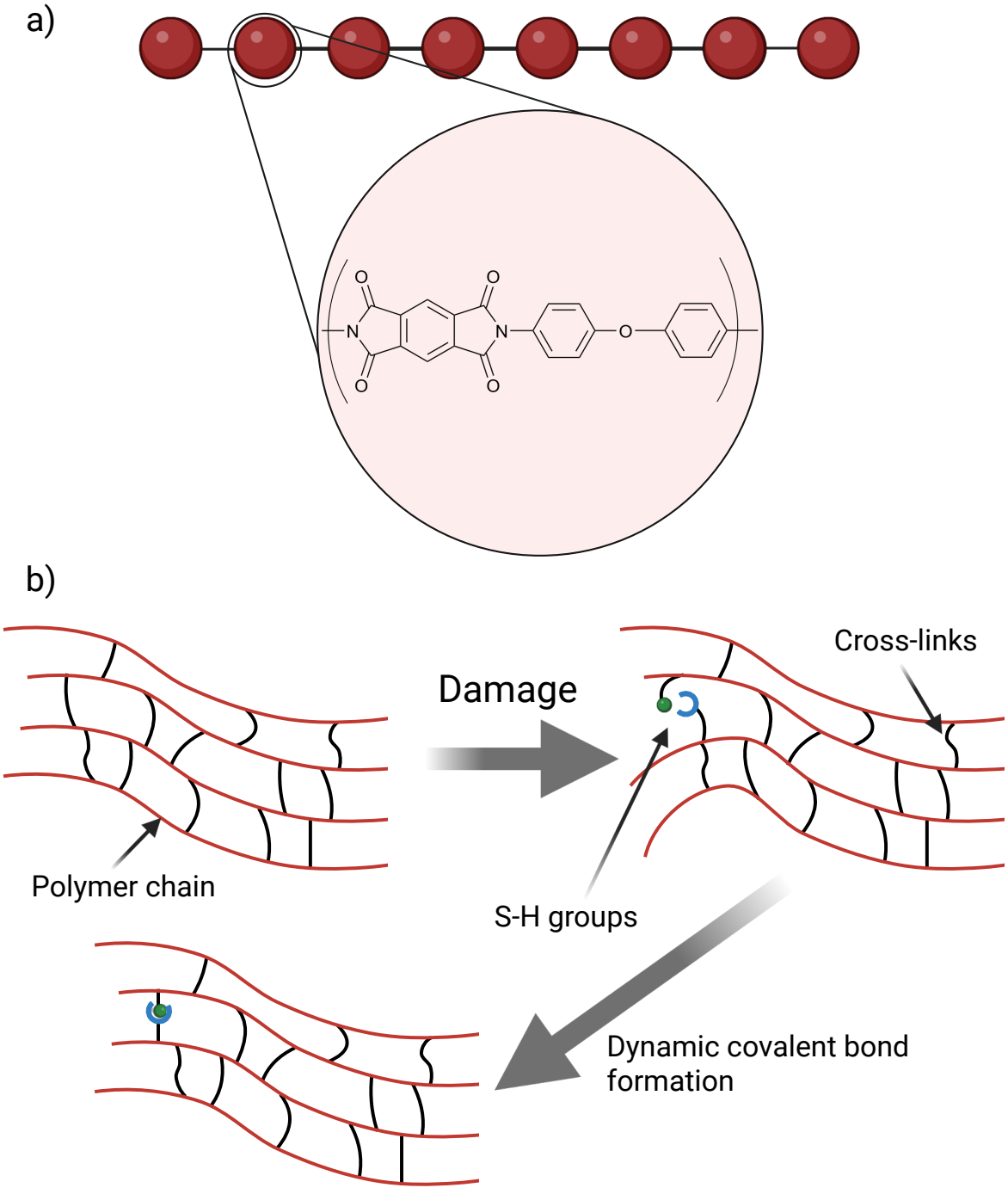


Figure 1.2: a) The polymer chain with a repeating chemical sub-unit. b) The mechanism of damage and repair for a self-healing polymer

process occurring without external input. For example, capsules that burst upon damage of the material and release a healing agent are classified as autonomous, whereas a photochemical reaction that needs UV light to activate[20,21] is a non-autonomous mechanism.

1.3 Comparison of Different Self-Healing Pathways

There are a variety of different pathways that can be taken to create a S-H polymer depending on the chemistry of the initial polymer. The pathways that will be focused on are intrinsically self-healed types, however, both autonomous and non-autonomous types will be considered.

The three pathways below rely on a property of some electrons called conjugation. Conjugation occurs when specific electron orbitals overlap in such as way that allows the electrons to move around the molecule. This is often exploited with a property of some atoms called electronegativity, having a highly electronegative atom in a molecule will attract conjugated electrons to create a local high density region of electrons and therefore a polarisation in a molecule.

Dynamic covalent bonds can form and break based on the conditions applied to them[22] and they are ideal for use as mechanisms for S-H due to their intrinsic nature and reliability in production of high yields.[23] This report investigates the use of two types of dynamic covalent bonding, Diels-Alder and Michael additions, and an example of a supramolecular interaction. Supramolecular chemistry is based on interactions that are non-covalent and may be more suited to a soft material.[24] This is due to the S-H mechanism of the cross-links that forms interactions therefore the polymer chains can move semi-independently.[24,25] Whereas, a dynamic covalent bond has a much higher bond energy[22] and can therefore get closer to the original physical properties of a polymer.

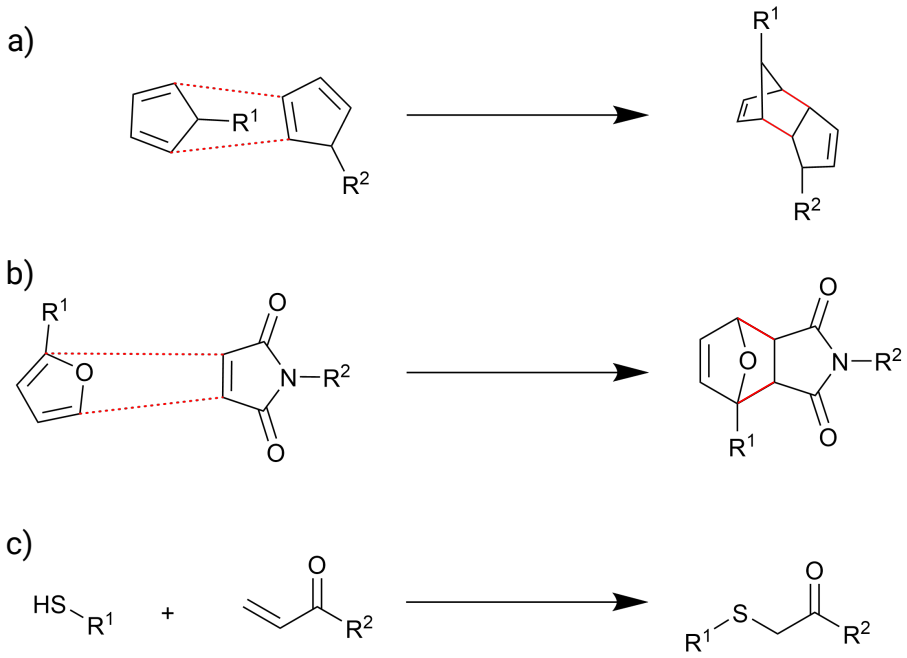


Figure 1.3: a) A Diels-Alder reaction between two molecules of cylopentadiene. b) A Diels-Alder reaction between a furan and maleimide. c) A Thiol-Michael Addition between a thiol and a conjugated alkene.

1.3.1 Using a Diels-Alder Addition as a Self-Healing Pathway

Diels-Alder (DA) reactions require a diene, a class of molecule characterised by having two sets of double bonds, and an alkene ("dieneophile").[22,26] The reaction occurs when conjugated electrons from the diene are free to move to form new bonds with the carbons either side of the alkene double bond, this creates a new ring structure. A faster reaction is generally observed with an electron rich diene and an electron poor dienophile.[27] The electronic character of the reagents can be altered by changing their local substituents. This class of reactions can have lots of different conditions depending on the molecules involved. Some variations can occur at room temperature,[28] which may make them an ideal choice for an autonomously healed medical device as the healing can occur whilst deployed in the body. However, many DA reactions are light[20,21] or heat activated.[29]

Some molecules such as cyclopentadiene can perform a DA reaction with itself, this has the potential to be an ideal choice for a S-H polymer as the molecule will never be in close proximity to a molecule that it cannot react with. However, some evidence has shown

that the bonds formed are not very strong and therefore high tensile strength materials may not be possible.[26]

One of the most common types of DA reaction is a heat activated reaction between a derivative of both furan and maleimide.[30] They react extremely accurately and require mild conditions,[26] therefore they are very suitable for application as a robotics material. One of the reasons that this reaction is so effective is that the maleimide has electron withdrawing carbonyl groups which make the dieneophile electron poor and therefore reacting faster. However, this mechanism has a disadvantage compared to cyclopentadiene as there are two halves of the end product. Therefore, there must be a two different processes occurring to create both the furan substituted and maleimide substituted chains. This increases the number of total steps in the reaction which increases the risk of low reaction yields and wasted material.

1.3.2 Using a Thiol-Michael Addition as a Self-Healing Pathway

Thiol Michael additions are a type of reaction that relies on a thiol, a molecule characterised by a sulfur containing group and a conjugated alkene.[31] The conjugated alkene used in a Michael addition is with a carbonyl group that will attract the electrons from the alkene to allow the thiol group to react. This has a similar advantage to the Diels-Alder reaction that it is not totally dependant on specific reactants merely groups of reactants. Therefore slight changes can be made to the adjoining groups to tweak the reaction conditions. This has been previously shown to have been successful with a heat activated reaction that could heal scratches after 15 to 30 minutes at 60 °C.[32]

1.3.3 Using a Pi-Stacking Approach as a Self-Healing Pathway

Pi-Stacking is a type of intermolecular interaction whereby conjugated systems interact based on the neighbouring substituents determining if a region is electron rich or poor.

These electron disparities cause an electrostatic interaction. This method commonly uses pyrenyl groups, these are polycyclic aromatic compounds, a very pi-electron rich species. These can be interacted with a napthalene diimide complex, an electron poor pi-system to give four pi-stacking interactions. A method has shown the creation of a polymer chain end group which can sandwich a napthalene group within two pyrenyl groups.[33] This method can be combined with other intermolecular interactions like hydrogen bonding[34] or mettalosupramolecular interactions to give a stronger interaction.[35]

1.4 Applications of Self-Healing Polymers in Soft Robotics

Continuing the theme of biological inspiration of soft robotics, S-H polymers can be used to extend the lifespan of components. In soft continuum robots, the goal is usually to make the robot softer than the environment surrounding it.[12] This presents a challenge that the environment can often be abrasive and can damage the device. It is therefore evident that a damage resilient material would be useful. Damage resilient materials can take multiple forms: Damage isolation, material elasticity, physical self healing such as low melting temperature materials and chemical self healing.[36]

1.4.1 State of the Art Self-Healing Soft Robots

State-of-the-art approaches using intrinsic S-H mechanisms are commonly based on DA chemistry.[18] These have been successfully implemented in tendon-driven[37] and pneumatically controlled soft robotic devices.[38,39] Such devices are capable of recovering from macroscopic damage, including punctures and cuts. Additionally, there are non-covalent approaches that utilize hydrogen bonding in the development of soft self-healing actuators, such as in McKibben muscle designs.[40]

1.4.2 Requirements for Self-Healing Polymers in Soft Robotics

Terryn *et al.*[18] states that there are 5 main criteria for a S-H polymer; macroscopic damage must be healed completely; healing can take place multiple times at the same location; there are minimal material property changes after healing; elastomeric behaviour is achieved; the reprocessing and recycling of a material is possible. Here we propose that elastomeric behaviours are non-essential and a more useful requirement would involve a timely S-H reaction.

1.5 Limitations with Current Materials for Soft Robotics

Damage control whilst a device is active can be a challenge,[41] this shows a need for autonomous S-H.[18] Non-autonomous S-H can be made "functionally autonomous" through the use of stimuli responsive mechanisms, however, the current options are slow in operation.[41] Currently recovery timescales of applied S-H polymers are around 25 hours.[36,39] By altering the polymer used or S-H pathway this could be improved. There is also a lack of an easy synthesis route for many of the current polymer options,[36] thereby increasing the cost and reducing widespread use. Also the number of damage-repair cycles in these devices are low, such as 3 or less.[39] Most devices do not achieve all 5 criteria laid out in the Terryn *et al.* review,[18] such as the McKibben design of A. Lopez Diaz *et al.*[40] only reports a healing efficiency of 65%.

Chapter 2

Project Aims and Objectives

Chapter 3

Synthesis of a Self-Healing Polymer

3.1 Results

3.1.1 Hydrolysis of the Polyimide

Our method was based on previous literature publications[2,42–48] where a alkaline solution is used to hydrolyse the polyimide to form a sodium salt intermediate and then an acidification to form the carboxlyic acid groups. We trialled the first step with a NaOH solution (conc.)[42–44] at room temperature for 90 mins, however when there was minimal change to the IR spectrum we increased the temperature to 120°C. Also we remade the NaOH solution to a concentration similar to[47] where we use 1 equiv. OH^- for every 1 monomer of the polyimide. This was chosen as to only partially hydrolyse one of the imide rings, to only expose one carboxylic acid group rather than the total hydrolysis to the tetrasubstituted benzene ring. However when looking at the IR spectrum of this we also observed minimal hydrolysis.

A widely accepted base-catalysed amide hydrolysis route is through a second order pathway, where the hydroxy group nucleophillically adds to the carbonyl group to form a tetrahedral intermediate[49] before this cleaves into two and the amine is eliminated. This

is also shown in pthalimide structures[50] when in solution.

However, there are changes when the reaction contains macromolecules, it has been shown in previous literature that there is a third order dependence on [OH^-] and therefore follows a mechanism that starts with the nucleophillic addition of the OH^- to form a charged intermediate that undergoes rate-determining cleavage of the C-N bond to form the polyamic acid.[42] There is also published results that propose that, due the hydrostatic repulsion, the rate of reaction is lowered when a polymeric chain has increasing numbers of negative charges,[51] this may mean that the formation of the monanion intermediate could hinder the reaction.

While multiple literature sources opt to use KOH,[2,45,47,48] we trialled NaOH. In retrospect, using KOH may have been a better approach as many KOH based literature sources have a good success rate. However, we attribute most of the failure in this experiment to the low concentration of NaOH, as sources have stated that a concentration below 0.5M was ineffective.[2] This may explain why our testing with 0.04M was also ineffective. Additionally, some of the failure may be caused by the lack of addition of a reagent to maintain constant ionic strength, which has been shown to assist hydrolysis with NaOH.[42]

Structural Analysis

Structural determination was conducted via Infrared spectroscopy. A comparison was made between the attempted hydrolysis of the sheets and a purchased form (polyamic acid).

Characteristic peaks in hydrolysis attempts are at around 1500cm^{-1} , this peak corresponds to the C=C bond in the central benzene ring[2,45] along with the 1235 cm^{-1} aromatic ether stretch[45] both which do not change throughout the hydrolysis. We also have the C=O symmetric stretching mode at around 1710 cm^{-1},[2,43,45–47] the peak around 1775 cm^{-1} as the symmetric C=O stretch[45–47] and the C-N-C imide ring stretch at 1370

cm^{-1},[45,46]all of which decrease in magnitude during hydrolysis.[2,46]

One literature source[52] states that the symmetric C=O stretch is at 1775 cm^{-1} and the asymmetric is at 1720 cm^{-1} however, due to the dispariuty between this and the other sources, we disregard this result.

Some literature estimate their degree of hydrolysis with a comparison of the ratio of intensities of the 1720:1500 cm^{-1} peaks,[2,47] these are summarised in the table 3.1. In our findings we see this peak slightly lower at around 1710 cm^{-1}.

As in the Wu 2017 paper,[2] we assumed the Kapton sample to have 100% imide content on the surface, which finds that the polyamic acid sample contains 18.71% surface imide content. We also find that the surface imide content decreases with both time and an increase in NaOH concentration. We find that the best of our hydrolysis attempts was the 90 minute, 0.2M hydrolysis at room temperature, giving a total imide surface content of 94.24%, meaning that 5.76% of the imide rings were cleaved.

Sample	**Base**	**Conc. /M**	$\mathbf{A_{1500}}$	$\mathbf{A_{1720}}$	$\mathbf{A_{1720}/A_{1500}}$	**Rel. imide content**
90 min, 25°C	NaOH	0.2	0.2243	0.2304	1.0273	0.9424
20 min, 120°C	NaOH	0.03	0.1062	0.1167	1.0990	1.0082
40 min, 120°C	NaOH	0.03	0.1608	0.1735	1.0789	0.9898
Kapton	—	—	0.1125	0.1226	1.0900	1.0000
PAA	—	—	0.8508	0.1735	0.2040	0.1871
Wu (2017)	KOH	0	0.0513	0.0587	1.1440	1.0000
Wu (2017)	KOH	0.1	0.0504	0.0573	1.1375	0.9943
Wu (2017)	KOH	0.5	0.0511	0.0568	1.1096	0.9699
Wu (2017)	KOH	1	0.0587	0.0617	1.0519	0.9194
Wu (2017)	KOH	5	0.0763	0.0072	0.0946	0.0827

Table 3.1: ATR-FTIR absorbance ratios and relative imide surface content for various samples in comparison to Wu (2017)[2]

3.1.2 Spectrophotometry

The test for carboxylic acids as laid out by GS Johar would in theory be a good test for determination of the hydrolysis of the polyamic acid. Therefore we tested for its interaction with acetic acid as the assumed positive test, along with DMSO as this was

the solvent the polyamic acid was dissolved in, the polyamic acid in DMSO, water and also hydrochloric acid to test the interaction was not pH based. This was tested with three different concentrations of dyes. Only the results from the least concentrated samples will be included as the absorbance is too high for most of the analysis on the more highly concentrated versions.

In the literature, upon mixing of the Eosin Y and Methylene blue "[there is an immediate colour change to green]". However, in our experience the colour was initially blue.

The absorbance of the sample with just water should give a baseline reading of how the absorbance of the dyes are. For a conclusive test, we should see the interaction between the dye mixture and the polyamic acid, should be similar. We see the flat region in the control test, between 600 and 700 nm grows in a similar shape when both acetic acid and PAA are added. However, there is also a similar shape from the sample containing DMSO. This indicates that the test may not be very effective. A repeat of this test could be performed with a less concentrated dye mixture and the PAA in a different solvent as most of the peaks in the PAA scan are in the DMSO. A wider range could also be used as there is high absorbance in the low region (below 300nmn) that is not visible.

note: although the paper states to use Eosin Y, there is a common misnaming and the Eosin Y disodium salt must be used.

From the UV-vis data, it can be determined that the GS Johar method is not appropriate for a determination of hydrolysis.

3.1.3 Cross-linking of the Polyimide

Cross-linking attempts follow a route where ethylene carbonate (EC) is used to form an ester with the carboxylic acids on neighbouring polymer chains, thereby joining them together with a "cross-link". This was attempted with three different salts ($Et_4NBr, ZnCl_2, FeCl_3$) as catalysts.

During the cross-linking reaction with Et_4NBr, an orange-brown suspension was formed that when washed, filtered and freeze dried gave a pale orange powder. With $ZnCl_2$, the liquid was a dark orange clear liquid that, when water was added, gave a single clump of grey polymeric residue. This was freeze dried to remove the excess water and then analysed. The cross-linking reaction with the $FeCl_3$, gave a dark orange/brown gel, which gave a grey gum-like texture in small pieces when water was added. This was also frozen and analysed.

During a cross-linking with ethylene carbonate, CO_2 is released. This was a concern as the monowave reactor has a pressure limit of 20 bar. To ensure there was a reliable safety buffer, we set an artificial limit of 5 bar. The ideal Gas law was used to ensure this limit was held.

$$PV = nRT \rightarrow n = \frac{PV}{RT}$$

P = 500 kPa, V = 10 mL, R = 8.314 J K^{-}1 mol^{-}1, T = 423K

$$n = \frac{500 \times 10}{8.314 \times 423} = 1.42$$

Using $mass = Mr \times n$, where Mr = 44, n = 1.42

$$Mr \times n = 62.48$$

Detailed mechanistic studies of the reaction between a carboxylic acid and ethylene carbonate in the presence of a quaternary ammonium salt have been performed in other literature.[53–55] Yoshino (1977) states that the mechanism begins with a nucleophillic ativation by the anion of the catalytic salt on a methylene group of the EC, this cleaves the hydrogen from the alcohol group of the carboxylic acid which can now nucleophillically substitute the halide to form the ester. Fagerburg (1984) disagree with this mechanism as they propose that the rate of the reaction would be dependent on the anion of the catalytic salt, however they did not find this experimentally. They found during a reaction of

the catalytic salt and EC alone, the activation energy of the EC decomposition was much higher than any esterification attempts. They therefore propose a mechanism involving the attack of a carboxylate species on the methylene carbon of the EC to decompose the ring and form the ester product. Given the breadth of Fagerburg's work, we consider this mechanism to be more robust than the one proposed by Yoshino.

The change of catalyst to a lewis acid ($ZnCl_2, FeCl_3$) was to encourage a direct activation of the carbonyl group on the EC, the mechanism for this proposed step was as follows: The lewis acid activates the carbonyl on the EC by forming a "complex" with it, this allows the alkoxide to attack the methylene group of the EC and form the hydroxyethyl ester.[56] Von Seggern (2020) however report, the use of zinc based catalysts was found to be ineffective in this reaction.[56]

Spectral analysis

Spectral analysis was performed mainly with IR, but also with ^{1}H NMR and DOSY.

From the IR data, the crosslinking with Et_4NBr was deemed to be unsuccessful as the product didn't have the key peaks of the C-O bond in the ester or any alcohol peak. The alcohol peaks would be likely as the crosslinking was unlikely to be 100% successful and there would likely be the single ester product formed. However, neither of these were present and since the product was insoluble, no NMR could be obtained. This IR spectrum was compared with the spectrum of Kapton (i.e. the ring closed form of the polyamic acid) and the similarities confirmed that the imide ring reformation had occured. Key spectral peaks confirming the lack of crosslinking could be at 1391 cm^{-1} where genrally the C-N-C peak is found a 1370 cm^{-1}, and the increase in the magnitude of the C=O stretch at 1715 cm^{-1}. These peaks would indicate the imide ring reforming as opposed to the esterification product.

From the IR data of the $ZnCl_2$ and the $FeCl_3$, there are two new peaks at around 1800 cm^{-1} which are not in either the Kapton or the polyamic acid. This shows the presence

of the starting material as the C=O bonds in liquid EC.[57] A new peak was also obsevered at 4.43ppm in the NMR spectrum, however, when adding 2 μL of EC in a "spike test", the peak increased in size, confirming that the new peak was starting material. Other peaks in the $ZnCl_2$ and $FeCl_3$ samples are indicative of unreacted polyamic acid. After increasing the temperature of the $ZnCl_2$ catalysed sample, the peaks positions show the intramolecular cyclisation of the imide ring with the C-N-C stretch at 1370cm^{-1}.

DOSY data was collected for the $ZnCl_2$ sample

3.1.4 Moulding of Polyamic Acid

The moulding without drying the polymer gave a yellow powder, this is likely due to the boiling point of the NMP being higher than the reported cure temperature of the polyamic acid. Therefore the polymer cured whilst still in solution and therefore didn't form a fully stable cure.

However, the attempt after a week of drying was much more successful and the top layer gave a glassy smooth layer whilst layers below were still not cured in one piece. This is likely due to the polyamic acid forming a skin on top of the mould. This is due to the way that the smartvap works, the smartvap pumps a vortex of air over the surface of the material. This means that the top layer dries out much faster than layers below. This is fine when the sample in solution is small, however polymeric materials behave differently.

3.1.5 Catalytic Imidisation of polyimide

During the attempted crosslinking reactions, we observed the imide ring formation at a lower temperature than in the literature. We attempted to recreate this with the polyamic acid and the $ZnCl_2$ after observing that the crosslinking attempts would form the kapton at lower temperatures than expected.

The reaction without the catalyst was as expected, the material got slightly darker, however we attribute this to some of the solvent evaporating and therefore the substrate becoming more concentrated. When comparing the spectrum to the polyamic acid on its own, we determined that [nothing had happened?]. However in comparison to the perparation with $ZnCl_2$, the IR spectrum showed a spectrum more similar to the kapton. The spectrum has extra peaks at (vals) which match the spectrum from the polyamic acid. This likely shows that the reaction was incomplete. Further purification and analysis is neccessary

As in our analysis against Wu 2017, we can assess the success of the imidisation by comparing the A1720 to the A1500:

table

3.2 Experimental

3.2.1 Equipment

The main piece of equipment used was the monowave reactor (details). This allows the user to seal a 10 mL vial to heat the sample under a reduced pressure and therefore to be heated above their boiling points. The monowave reactor requires a specific tube that can be sealed with a rubber cap and a PTFE septum, the monowave contains a magnetic stir element and therefore an oval magnetic stir bar was included.

3.2.2 The Hydrolysis of a Polyimide

Polyoxydiphenylene mellitimide (Kapton) film (0.2g, thickness of x) was cut into pieces of approx. $3mm^2$ and placed into a reaction tube along with a solution of NaOH (). This tube was placed into a monowave reactor that was set to ramp for 20 minutes and hold

Dye Mix	Parts water	Parts Eosin Y (0.5%)	Parts Methylene Blue (0.1%)
Mix 1	10	2	1
Mix 2	5	2	1
Mix 3	0	2	1

Table 3.2: table caption

Sample number	Test compound	Dye mix
S01	Dimethyl Sulfoxide	Dye 1
S02	Polyamic Acid solution	Dye 1
S03	Water	Dye 1
S04	Acetic Acid	Dye 1
S05	Hydrochloric Acid	Dye 1
S06	Dimethyl Sulfoxide	Dye 2
S07	Polyamic Acid solution	Dye 2
S08	Water	Dye 2
S09	Acetic Acid	Dye 2
S10	Hydrochloric Acid	Dye 2
S11	Dimethyl Sulfoxide	Dye 3
S12	Polyamic Acid Solution	Dye 3
S13	Water	Dye 3
S14	Acetic Acid	Dye 3
S15	Hydrochloric Acid	Dye 3

Table 3.3: table caption

for 20 minutes at 100°C. There was no observable difference, so a sample was taken and the remaining mixture was placed onto the monowave under the same conditions. The mixture was filtered and thoroughly washed with water before IR was collected.

3.2.3 Spectrophotometry as a Determination Method for Hydrolysis

The preparation of samples for the spectrophotometric analysis was as follows:

The dye mixes were replicated from previous literature: "3 mL 0.5% Eosin Y solution was added to 1.5mL 0.1% Methylene Blue solution" [cite GS Johar]. This solution was diluted with 10 parts water and 5 parts water to give three dye mixes (summarised in 3.2)

Catalytic Salt	Mass of Salt /g	Reaction Temperature /°C	Analytical Method
Et_4NBr	0.08	150	IR
$FeCl_3$	0.02	100	IR, NMR
$ZnCl_2$	0.03	100	IR, NMR, DOSY

Table 3.4: table caption

Test compound (2.7mL) was added to a sample vial along with one of the dye mixes (0.3 mL) to form samples summarised in 3.3. These were inverted to ensure mixing before being transferred to a cuvette and placed in the spectrophotometer to test between 250 and 750nm.

3.2.4 Crosslinking of Polyamic acid

This procedure outlines the crosslinking procedure of Polyamic acid via ethylene carbonate using a catalytic salt as outlined in previously published procedures.

A mixture of polyamic acid (0.51mL, conc) was placed into a monowave tube with ethylene carbonate (0.08g) and a catalytic salt. These were placed into the monowave to ramp for 20 minutes and hold at the reaction temperature for an hour. The products were washed with water and filtered before being freeze dried and "run" on the spectrometers. The reagent amounts and processing is summarised in table 3.4.

3.2.5 Crosslinking Spectral Data

tetraethylammonium bromide Crosslinking

A orange powder (grams,moles,yield) was isolated, ν max/cm^{-1}(solid);

Iron chloride Crosslinking

A white gum (grams,moles,yield) was isolated, δ H(400MHz, DMSO); , δ C(100MHz, DMSO); , ν max/cm^{-1}(solid);

Zinc chloride Crosslinking

Brittle brown pieces (grams,moles,yield) were isolated, δ H(400MHz, DMSO); , δ C(100MHz, DMSO); , ν max/cm^{-1}(solid);

3.2.6 Catalytic Imidisation of Polyamic Acid

Polyamic acid (0.51mL), $ZnCl_2$ (0.02g) and a magnetic stirbar were added to a Monowave tube. This was placed in the monowave at 110°C for 1 hour (ramp for 20 mins, hold for 60 mins). A control experiment without the $ZnCl_2$ was also placed under the same conditions. The control expeiment gave a clear dark brown viscous liquid which when dried gave a gum-like pale orange solid. The test with the $ZnCl_2$ gave a similar dark brown liquid however this was slightly cloudy and when processed, gave a orange powder.

Control Spectrum

IR(solid): vals, vals cm^{-1}

Catalytic Imidisation Spectrum

IR(solid): vals, vals cm^{-1}

Chapter 4

Construction of a Bellows Based Actuator

4.1 Design

Actuators were constructed using a combination of a folded polyimide sleeve and a internal multilumen silicon tube with nitinol wires for actuation. Our methodology is inspired by similar "bellows-style" approaches to construct continuum devices (references) with the key difference in the use of Polyimide sheets (Kapton) forming the main body of the device.

This work focuses on the use of Kapton (thickness =) due to its good robustness and high temperature capabilities.

4.1.1 Construction of a Folded Protective Sleeve

The polyimide film was folded into a []-style design as based on the previously published work [reference]. To achieve the folding a 2D drawing of the fold lines was drawn in Fusion360 and printed onto paper. This paper was taped to the table and a sheet of

Kapton was taped over the top, the Kapton is clear so the printed lines are visible through. The lines were then traced onto the Kapton with a ruler and a ballpoint pen. The ballpoint pen marked the desired fold lines and also partially indented on the lines. The sheet of kapton was removed from the table and the outer edge was cut with scissors. The Kapton sheet was then folded along the lines as shown in the figure [ref figure]. The folds on the sheet were puished back out and the kapton was rolled into a cylinder. This was taped along the overlapped edge to form a temporary seal while maintaining care to line up the folds. The tape was removed in small sections and JBweld HighHeat was applied until the whole adhesion region. This was threaded onto a plastic tube with a diameter approx. 2mm smaller than the kapton sleeve. The sleeve was taped to the tube and the adhesion region was taped over the top to clamp the glued region. This was left for 24 hours to cure. The glueing and the curing process was performed in a Fume cupboard [brand/details] due to the risk of respiratory irritation through the curing process. After curing, the sleeve was removed from the plastic tube and a strip of Kapton tape was placed over the edge on the adhesion region. This improved the flexiblity of the region. one of the ends was folded over to make a cap and this was also glued. In this work, this assembly is known as the "Kapton sleeve" or "polyimide sleeve".

Manual folding versus using a plotter

A commercially available plotter (Cricut®) was used with a scribe tool and a cutter to achieve a rapid manufacturing of a folded sleeve. This was attempted with paper initally for cheap prototyping and then with the polymer film. This would avoid the time consuming folding steps

The Cricut was set up with the cricut design space software. This software allows the user to draw and communicate with the cricut either through bluetooth or via cable. The cricut has an adhesive cutting mat to hold the material in place whilst the tool interacts with it. For our testing, we drew a rectangle (10 x 5 cm) and set the cricut to cut on this line. Then a line (5cm) was drawn to split the long edge into two squares ($5cm^2$), this

line was set to score. These two drawings were attached so as they would be performed in the same material loading cycle, the scoring was set to be first in the cricut's "job queue" to avoid there being potential for the score tool to catch the newly cut edge of the polymer and causee the adhesion between the sheet and the polymer to fail. A piece of the desired material was adhered to the cutting mat and the cricut was started. This method was repeated for 2 thicknesses of Kapton and 80gsm paper (roughly 0.25mm).

The three pieces of cut material were taped by the top edge to the jaws of a vice. The vice had the jaws enlarged by 3D printed pieces to ensure that the clamping would occur across the whole of the material. The vice was manually controlled to compress the pieces of material into the crease to form a fold and then slowly decompress to evaluate the fold. The folding with the paper was highly effective, when the jaws were opened the paper was clearly separated into two sections by a fold in the centre. Both thicknesses of Kapton were not so effective, when they were decompressed there was no clear fold line and the material was left haning in a curve. The plotter can visible score the Kapton, however, the score lines do not define a crease as the scoring tool does with paper. This phenomenon is explored further in the [put paper here], where polyimide is shown to not form precise folds.

We determine here that the use of a plotter could be useful in the rapid production of folded sleeves, however in its current form, the cricut is not a viable route for creasing Kapton. A comprimise could be to use the cricut to draw the fold lines by using a pen in place of the scribe tool, this would allow the user to bypass the printing and tracing the pattern by hand.

4.1.2 Complete Assembly

The complete assembly was formed with the polyimide sleeve, the silicon continuum device and 3D printed parts.

The silicon continuum device contains a silicon 7-hole multilumen tube with a 3D printed

cap, the lumen are arranged so that there are 6 equally spaced edge lumen and one central lumen. There are 3 nitinol wires that are approximately double the length of the silicon tube and these are threaded through one of the edge lumen, through the 3D printed cap and back through a neighbouring lumen. This is repeated another 2 times until all 6 outer lumen contain a niinol wire. We define the disatl end as the end with the 3D printed cap. At the proximal end, where the ends of the nitinol wires are, these are fed into 3 3D printed handles where the wires are held with 1.5mm grub screws. This assembly is threaded through a stiffer tube (approx. 15mm diameter) and a cable gland on the end of this stiff tube holds the silicon tube and the larger tube together. The device can be actuater by fixing the stiff tube in place and pulling on the 3D printed handles. In some instance, we increased the stiffness of the device to improve the controllability of the deivce by putting a nitinol wire through the centre lumen. This centre lumen can also be used for adding to the functionality of the device by presenting an opportunity for targeted drug delivery, a tool channel or for wires from tip-based components.

The construction method for the completed assembly is as follows: the silicon continuum device is threaded through the middle. The glued end of the kapton sleeve has a cable gland (size and details) and a 3D printed octagonal connecting piece glued to it. the silicon device is threaded through the middle of the sleeve and secured in place with the cable gland. The other end of the sleeve is clamped to the silicon continuum device with 3D printed parts to form the completed device.

4.2 Methodology

4.2.1 Stiffness Characterisation

THe stiffness of the polyimide sleeve applies an additional force on the silicon actuation device. For instance, when the device is pressurised, the additional forces applied to the device from the stiffness of the sleeve will affect the amount of air pressure needed to

vary the actuators length. To characterise the stiffness profile, a linear stage was used with a load cell (brand) held in a SmallRig Magic Arm. This was attached to the end of the polyimide sleeve which was clamped in a 3D printed stand. The whole experimental setup was attached to a ThorLabs Breadboard. The linear stage was set to 60mm (the resting length of the actuator was 120mm, the linear stage is in reverse so to get to 60mm, the linear displacement is 90) and the load cell was zeroed. The load cell was moved to 150mm (actuator length is 20mm compressed) and then the linear stage was set to go to 0mm at 0.5mm s^{-1}. This contained a compression force and a stretching force. The speed of the actuator was translated into the displacement using:

$$v = \frac{d}{t}$$

. The load cell data was collected using DSCUSB Toolkit and processed in python.

4.2.2 Temperature Stability of the Polyimide Sleeve

Using the same experimental set-up as the stiffness characterisation. The kapton® sleeve was placed in an oven at 200°C for 1 hour, left to cool and then compressed and stretched as before.

4.2.3 Workspace Characterisation

To characterise the usable workspace of the device, an experiment was set-up with infrared cameras (optitrack) to track the tip position of the device. When assembled, the device was not very predictably controllable partially due to the additional weight at the tip of the added cable gland. This meant that the actuator

4.2.4 Moulding of polyamic acid

Polyamic acid undergoes an imidisation reaction at 150-200° to form Kapton. It shoudl, in theory be prossible to pour the polyamic acid into a mould and heat to form Kapton. This was attempted by pouring the PAA into a mould and heating to 150°C for 20 mins to cause imidisation. This was repeated after placing half the mould into the smartvap with a 3D printed adaptor. The smartvap heats the sample and blows a stream of air into a vortex over a sample to remove the solvent. This was performed for a week before then heating the sample to 150°C for 1 hour.

4.2.5 Moulding of Polyamic Acid

Polyamic acid in NMP (stats) was poured into a Cone-shaped mould. This was assembled with a 3D printed part to connect to the SmartVap (Supplier) for 1 week at 100°C. This reduced the volume and made the liquid a darker brown. This was heated to 150°C for 1 hour to cause imidisation before being extracted from the mould with a scalpel. This experiment was also repeated without drying off the solvent first.

4.3 Results

4.3.1 Stiffness Characterisation

As the force from compression and extension of the actuator was obtained on one experimental set up, the force in extension is a negative force. This was only obtained for 60mm paste the relaxed lenght as, extension further than this made folds on the actuator invert and reversible retraction was not possible.

The load cell was zeroed at 90mm displacement, however when the actuator reached 90mm in length the force measured was -0.5N. We believe this is due to the folding in

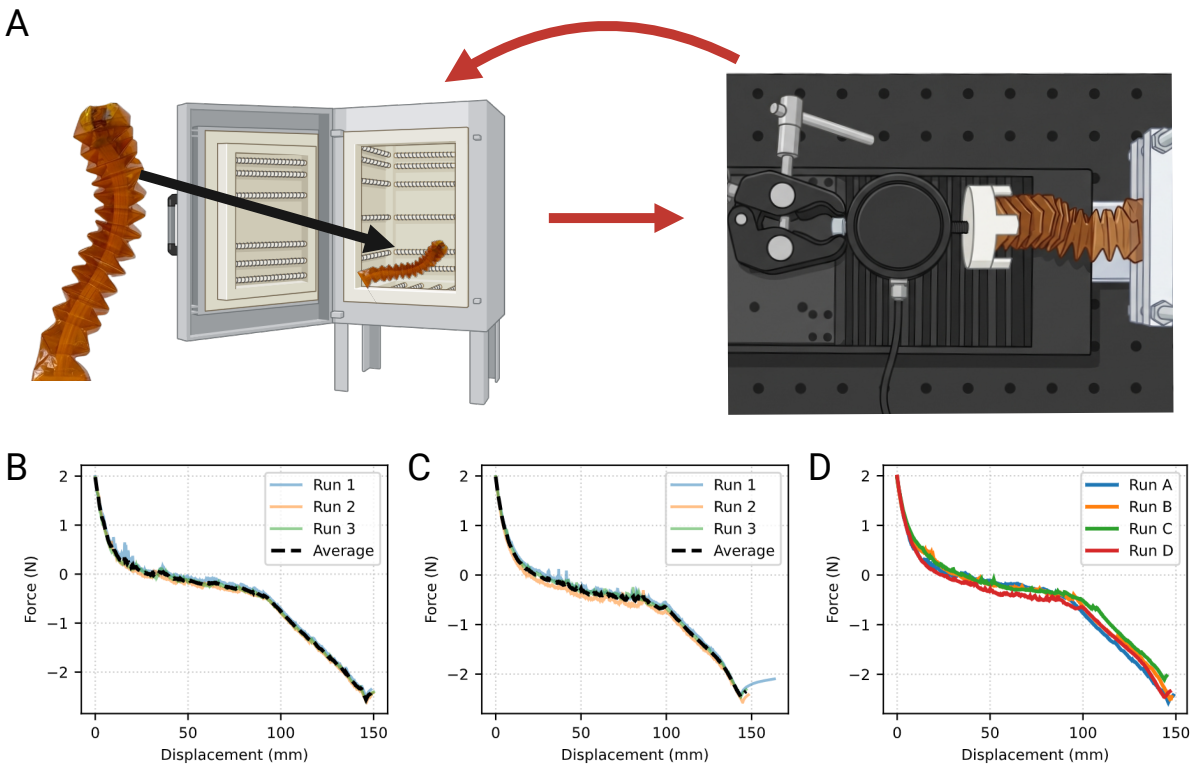


Figure 4.1: A) The Kapton sleeve is placed into an oven for an hour and then placed onto the stiffness characterisation experimental setup. B) The initial profile of the actuator before heating. C) The profile after 3 cycles in the oven. D) A plot comparing the average force vs. displacement across all cycles

the sleeve having some elastic characteristic [cite the kapton folding paper] and therefore taking up the folds and making the relaxed length temporarily shorter.

The maximal force under compression was (x) and the tensile force was (y). during compression it appears that an exponential would fit and during extension the line is linear.

4.3.2 Temperature Stability of the Bellows

To evaluate the stability of the acutator, tis was placed in the oven. After removing and running the sample 3 times after an hour each time. We determined that the actuator was very temperatrue stable.

Data from te he load cell was processed in python and plotted in graphs. The plots were

trimmed to start once the force was over 0.2N. This gave plots as shown in 4.1.

4.3.3 Workspace Characterisation

To characteri the usable workspace of the device, a

4.3.4 Navigation

To demonstrate the robots ability to navigate an arbitrary path, a route was set up using 3D printed obstacles to simulate a path that a real world use case could need. The goal of the path was to show that fluid could be delivered through a channel in the silicon inner tube.

This path started shorted than the relaxed length of the actuator and so required the operator to pull on the silicon tube to retract the device before steering the device using the tendons.

Extending the device further than the relaxed length was difficult due to the soft the silicon tube buckling under compression. This meant that the maximum path length was limited to around 120mm.

The introduction of a pressurised approach to controlling the locomotion of the device is beyond the scope of this

4.3.5 Clinical Case Study: Drug Delivery in Stomach

Finally, we demonstrate the use of the bellows based actuator in a stomach training simulator for the application of targeted drug delivery. Here we use the same assembly as the previous navigations with the addition of water to simulate the drug delivery. We use an external camera (camera details) inserted through a separate entry into the stomach

to give an approximate location for completing the navigations.

Bibliography

[1] Matteo Cianchetti, Tommaso Ranzani, Giada Gerboni, Thrishantha Nanayakkara, Kaspar Althoefer, Prokar Dasgupta, and Arianna Menciassi. Soft robotics technologies to address shortcomings in today's minimally invasive surgery: the stiff-flop approach. *Soft Robotics*, **1**(2):122–131, 2014. [17/10/2025]. Available from: 10.1089/soro.2014.0001.

[2] Pei-Yu Wu, Ching-Hsuan Lin, and Chih-Ming Chen. Study of surface metallization of polyimide film and interfacial characterization. *Metals*, 7(6):189, 2017.

[3] Kevin M Lynch and Frank C Park. *Modern robotics*. Cambridge University Press, 2017. [01/10/2025]. Available from: 10.1017/9781316661239.

[4] Elliot W Hawkes, Laura H Blumenschein, Joseph D Greer, and Allison M Okamura. A soft robot that navigates its environment through growth. *Science Robotics*, **2**(8):eaan3028, 2017. [16/10/2025]. Available from: 10.1126/scirobotics.aan3028.

[5] Joshua Davy, Thomas P Dean, Nikita J Greenidge, Benjamin Calmé, Peter Lloyd, James H Chandler, and Pietro Valdastri. Magnetic fluid-driven vine robots for minimally invasive tissue biopsy sampling. *Advanced Intelligent Systems*, **7**:2400827, 2025. [16/10/2025]. Available from: 10.1002/aisy.202400827.

[6] VV Anderson. Tenser arm manipulator design. *ASME Trans.*, 67:1–12, 1967. [10/10/2025].

[7] Peter Lloyd, Ali Kafash Hoshiar, Tomas Da Veiga, Aleks Attanasio, Nils Marahrens, James Henry Chandler, and Pietro Valdastri. A learnt approach for the de-

sign of magnetically actuated shape forming soft tentacle robots. *IEEE Robotics and Automation Letters*, **5**(3):3937–3944, 2020. [16/10/2025]. Available from: 10.1109/LRA.2020.2983704.

[8] Filip Ilievski, Aaron D Mazzeo, Robert F Shepherd, Xin Chen, and George M Whitesides. Soft robotics for chemists. *Angewandte Chemie International Edition*, **50**(8):1890–1895, 2011. [01/10/2025]. Available from: 10.1002/ange.201006464.

[9] Jianglong Guo, Khaled Elgeneidy, Chaoqun Xiang, Niels Lohse, Laura Justham, and Jonathan Rossiter. Soft pneumatic grippers embedded with stretchable electroadhesion. *Smart Materials and Structures*, **27**(5):055006, 2018. [17/10/2025]. Available from: 10.1088/1361-665X/aab579.

[10] Matteo Russo, Seyed Mohammad Hadi Sadati, Xin Dong, Abdelkhalick Mohammad, Ian D Walker, Christos Bergeles, Kai Xu, and Dragos A Axinte. Continuum robots: An overview. *Advanced Intelligent Systems*, **5**(5):2200367, 2023. [02/10/2025]. Available from: 10.1002/aisy.202200367.

[11] Joshua Davy, Nikita J Greenidge, Sukjun Kim, Luke J Tinsley, Peter Lloyd, James H Chandler, Russell A Harris, Tania K Morimoto, and Pietro Valdastri. Vine robots with magnetic skin for surgical navigations. *IEEE Robotics and Automation Letters*, **9**(8):6888–6895, 2024. [16/10/2025]. Available from: 10.1109/LRA.2024.3412637.

[12] Daniela Rus and Michael T Tolley. Design, fabrication and control of soft robots. *Nature*, **521**(7553):467–475, 2015. [02/10/2025]. Available from: 10.1038/nature14543.

[13] Haider Abidi and Matteo Cianchetti. On intrinsic safety of soft robots. *Frontiers in Robotics and AI*, **4**:5, 2017. [10/10/2025]. Available from: 10.3389/frobt.2017.00005.

[14] Xinyu Guo, Wenbo Li, Fuyi Fang, Huyue Chen, Linchuan Zhao, Xiaoyong Fang, Zhiran Yi, Lei Shao, Guang Meng, and Wenming Zhang. Encoded sewing soft textile robots. *Science Advances*, **10**(1):eadk3855, 2024. [17/10/2025]. Available from: 10.1126/sciadv.adk3855.

[15] Thomas P Dean, Joshua Davy, Jane Scott, Pietro Valdastri, and James H Chandler. Anisotropic 3d-knitted sleeves as inverse pneumatic artificial muscles for soft and wearable robotics. In *2025 IEEE 8th International Conference on Soft Robotics (RoboSoft)*, pages 1–6. IEEE, 2025. [17/10/2025]. Available from: 10.1109/RoboSoft63089.2025.11020963.

[16] Ruchir Shanker Srivastav and Aarti P More. A comprehensive review of self-healing polymers: Mechanisms, types, and industry implications. *Polymers for Advanced Technologies*, **36**(2):e70092, 2025. [10/10/2025]. Available from: 10.1002/pat.70092.

[17] Wolfgang H Binder. *Self-healing polymers: from principles to applications*. John Wiley & Sons, 2013.

[18] Seppe Terryn, Jakob Langenbach, Ellen Roels, Joost Brancart, Camille Bakkali-Hassani, Quentin-Arthur Poutrel, Antonia Georgopoulou, Thomas George Thuruthel, Ali Safaei, Pasquale Ferrentino, et al. A review on self-healing polymers for soft robotics. *Materials Today*, **47**:pp.187–205, 2021. [01/10/2025]. Available from: 10.1016/j.mattod.2021.01.009.

[19] Scott R White, Nancy R Sottos, Philippe H Geubelle, Jeffrey S Moore, Michael R Kessler, SR Sriram, Eric N Brown, and S Viswanathan. Autonomic healing of polymer composites. *Nature*, 409(6822):794–797, 2001.

[20] Scott R Trenor, Allan R Shultz, Brian J Love, and Timothy E Long. Coumarins in polymers: from light harvesting to photo-cross-linkable tissue scaffolds. *Chemical reviews*, 104(6):3059–3078, 2004.

[21] Pablo Froimowicz, Daniel Klinger, and Katharina Landfester. Photoreactive nanoparticles as nanometric building blocks for the generation of self-healing hydrogel thin films. *Chemistry–A European Journal*, 17(44):12465–12475, 2011.

[22] Jie Guo. *Diels-Alder chemistry: An old workhorse for new self-healing nanocomposites and adhesives*. PhD thesis, University of Groningen, 2025. [12/10/2025]. Available from: 10.33612/diss.1198671697.

[23] Kyriacos Costa Nicolaou, Scott A Snyder, Tamsyn Montagnon, and Georgios Vassilikogiannakis. The diels–alder reaction in total synthesis. *Angewandte Chemie International Edition*, 41(10):1668–1698, 2002.

[24] Ying Yang and Marek W Urban. Self-healing of polymers via supramolecular chemistry. *Advanced Materials Interfaces*, **5**(17):1800384, 2018. [12/10/2025]. Available from: 10.1002/admi.201800384.

[25] Shiwanka V Wanasinghe, Obed J Dodo, and Dominik Konkolewicz. Dynamic bonds: adaptable timescales for responsive materials. *Angewandte Chemie*, 134(50):e202206938, 2022.

[26] Chirag R Ratwani, Ali Reza Kamali, and Amr M Abdelkader. Self-healing by diels-alder cycloaddition in advanced functional polymers: A review. *Progress in Materials Science*, **131**:101001, 2023. [02/10/2025]. Available from: 10.1016/j.pmatsci.2022.101001.

[27] Benoit Briou, Bruno Améduri, and Bernard Boutevin. Trends in the diels–alder reaction in polymer chemistry. *Chemical Society Reviews*, **50**(19):11055–11097, 2021. [01/10/2025]. Available from: 10.1039/D0CS01382J.

[28] David Huertas, Melinda Florscher, and Veljko Dragojlovic. Solvent-free diels–alder reactions of in situ generated cyclopentadiene. *Green Chemistry*, 11(1):91–95, 2009.

[29] Jinhui Li, Guoping Zhang, Libo Deng, Songfang Zhao, Yongju Gao, Kun Jiang, Rong Sun, and Chingping Wong. In situ polymerization of mechanically reinforced, thermally healable graphene oxide/polyurethane composites based on diels–alder chemistry. *Journal of Materials Chemistry A*, 2(48):20642–20649, 2014.

[30] Bryan Ingoglia. Microencapsulation of diels-alder monomers in self-healing materials. *Kaleidoscope*, 11(1):43, 2014.

[31] Devatha P Nair, Maciej Podgorski, Shunsuke Chatani, Tao Gong, Weixian Xi, Christopher R Fenoli, and Christopher N Bowman. The thiol-michael addition click

reaction: a powerful and widely used tool in materials chemistry. *Chemistry of Materials*, **26**(1):724–744, 2014. [06/10/2025]. Available from: 10.1021/cm402180t.

[32] Natascha Kuhl, Robert Geitner, Ranjita K Bose, Stefan Bode, Benjamin Dietzek, Michael Schmitt, Jürgen Popp, Santiago J Garcia, Sybrand van der Zwaag, Ulrich S Schubert, et al. Self-healing polymer networks based on reversible michael addition reactions. *Macromolecular Chemistry and Physics*, **217**(22):2541–2550, 2016. [10/10/2025].
Available from: 10.1002/macp.201600353.

[33] Stefano Burattini, Barnaby W Greenland, Wayne Hayes, Michael E Mackay, Stuart J Rowan, and Howard M Colquhoun. A supramolecular polymer based on tweezer-type π–π stacking interactions: Molecular design for healability and enhanced toughness. *Chemistry of Materials*, **23**(1):6–8, 2011. [15/10/2025]. Available from: 10.1021/cm102963k.

[34] Xin Yuan, Xiangyu Lin, Fuhao Dong, Xujuan Huang, He Liu, and Xu Xu. Self-healed, tough, and highly resilient elastomer facilitated by cooperative hydrogen-bonding interaction and π–π stacking interaction. *ACS Applied Polymer Materials*, **7**(3):1328–1337, 2025. [16/10/2025]. Available from: 10.1021/acsapm.4c03053.

[35] Jin-Feng Mei, Xiao-Yong Jia, Jian-Cheng Lai, Yang Sun, Cheng-Hui Li, Jun-Hua Wu, Yi Cao, Xiao-Zeng You, and Zhenan Bao. A highly stretchable and autonomous self-healing polymer based on combination of pt···pt and π–π interactions. *Macromolecular Rapid Communications*, **37**(20):1667–1675, 2016. [15/10/2025]. Available from: 10.1002/marc.201600428.

[36] R Adam Bilodeau and Rebecca K Kramer. Self-healing and damage resilience for soft robotics: A review. *Frontiers in Robotics and AI*, **4**:48, 2017. [01/10/2025]. Available from: 10.3389/frobt.2017.00048.

[37] Ellen Roels, Seppe Terryn, Joost Brancart, Guy Van Assche, and Bram Vanderborght. A multi-material self-healing soft gripper. In *2019 2nd IEEE International*

Conference on Soft Robotics (RoboSoft), pages 316–321. IEEE, 2019. [16/10/2025]. Available from: 10.1109/ROBOSOFT.2019.8722781.

[38] Seppe Terryn, Joost Brancart, Ellen Roels, Guy Van Assche, and Bram Vanderborght. Room temperature self-healing in soft pneumatic robotics: Autonomous self-healing in a diels–alder polymer network. *IEEE Robotics & Automation Magazine*, **27**(4):44–55, 2020. [16/10/2025].
Available from: 10.1109/MRA.2020.3024275.

[39] Seppe Terryn, Joost Brancart, Dirk Lefeber, Guy Van Assche, and Bram Vanderborght. A pneumatic artificial muscle manufactured out of self-healing polymers that can repair macroscopic damages. *IEEE Robotics and Automation Letters*, **3**(1):16–21, 2017. [17/10/2025].
Available from: 10.1109/LRA.2017.2724140.

[40] Antonio López-Díaz, Ana Martín-Pacheco, Alicia Naranjo, Cristina Martín, M Antonia Herrero, Ester Vázquez, and Andrés S Vázquez. Autonomous self-healing pneumatic mckibben muscle based on a new hydrogel material. In *2020 3rd IEEE International Conference on Soft Robotics (RoboSoft)*, pages 13–18. IEEE, 2020. [16/10/2025]. Available from: 10.1109/RoboSoft48309.2020.9115992.

[41] Seyedreza Kashef Tabrizian, Seppe Terryn, and Bram Vanderborght. Toward autonomous self-healing in soft robotics: A review and perspective for future research. *Advanced Intelligent Systems*, **7**(8):2400790, 2025. [16/10/2025]. Available from: 10.1002/aisy.202400790.

[42] Lori E Stephans, Anthony Myles, and Richard R Thomas. Kinetics of alkaline hydrolysis of a polyimide surface. *Langmuir*, 16(10):4706–4710, 2000.

[43] Richard R Thomas, Stephen L Buchwalter, L Paivikki Buchwalter, and Taina H Chao. Organic chemistry on a polyimide surface. *Macromolecules*, 25(18):4559–4568, 1992.

[44] Miro M Plechaty and Richard R Thomas. Chemical surface-modification of a polyimide: Characterization and a kinetic study on the diffusion of ions in a modified polymer. *Journal of the Electrochemical Society*, 139(3):810–821, 1992.

[45] Wei-Ping Dow, Guo-Liang Liao, Shang-En Huang, and Sinn-Wen Chen. Modification of cu nanoparticles with a disulfide for polyimide metallization. *Journal of Materials Chemistry*, 20(18):3600–3609, 2010.

[46] Tzu-Jung Liu, Chien-Han Chen, Pei-Yu Wu, Ching-Hsuan Lin, and Chih-Ming Chen. Efficient and adhesiveless metallization of flexible polyimide by functional grafting of carboxylic acid groups. *Langmuir*, 35(22):7212–7221, 2019.

[47] G Xue. Studies of etching mechanism of polyimide films on silicon chips. *Die Angewandte Makromolekulare Chemie: Applied Macromolecular Chemistry and Physics*, 142(1):61–68, 1986.

[48] Nancy C Stoffel, Meng Hsieh, Sweta Chandra, and Edward J Kramer. Surface modification studies of polyimide films using rutherford backscattering and forward recoil spectrometry. *Chemistry of materials*, 8(5):1035–1041, 1996.

[49] Charmian O'Connor. Acidic and basic amide hydrolysis. *Quarterly Reviews, Chemical Society*, 24(4):553–564, 1970.

[50] M Niyaz Khan. Kinetics and mechanism of the alkaline hydrolysis of maleimide. *Journal of pharmaceutical sciences*, 73(12):1767–1771, 1984.

[51] J Moens and G Smets. Alkaline and acid hydrolysis of polyvinylamides. *Journal of Polymer Science*, 23(104):931–948, 1957.

[52] Jin-Hae Chang, Kwang Min Park, Soo-Min Lee, and Jae Buem Oh. Two-step thermal conversion from poly (amic acid) to polybenzoxazole via polyimide: Their thermal and mechanical properties. *Journal of Polymer Science Part B: Polymer Physics*, 38(19):2537–2545, 2000.

[53] Teruo Yoshino, Shigeru Inaba, Hajime Komura, and Yoshiharu Ishido. Synthetic studies with carbonates. part 6. syntheses of 2-hydroxyethyl derivatives by reactions of ethylene carbonate with carboxylic acids or heterocycles in the presence of tetraethylammonium halides or under autocatalytic conditions. *Journal of the Chemical Society, Perkin Transactions 1*, 1(11):1266–1272, 1977.

[54] David R Fagerburg. Reaction kinetics of the model reaction of benzoic acid with ethylene carbonate. *Journal of Macromolecular Science—Chemistry*, 21(2):141–154, 1984.

[55] David R Fagerburg. Reaction kinetics of the reaction of terephthalic acid and ethylene carbonates. *Journal of applied polymer science*, 30(9):3617–3623, 1985.

[56] Nils von Seggern, Tamara Schindler, and Stefan Naumann. Dual catalytic ring-opening polymerization of ethylene carbonate for the preparation of degradable peg. *Biomacromolecules*, 21(7):2661–2669, 2020.

[57] Bruno Fortunato, Paolo Mirone, and Glancaelo Fini. Infrared and raman spectra and vibrational assignment of ethylene carbonate. *Spectrochimica Acta Part A: Molecular Spectroscopy*, 27(9):1917–1927, 1971.